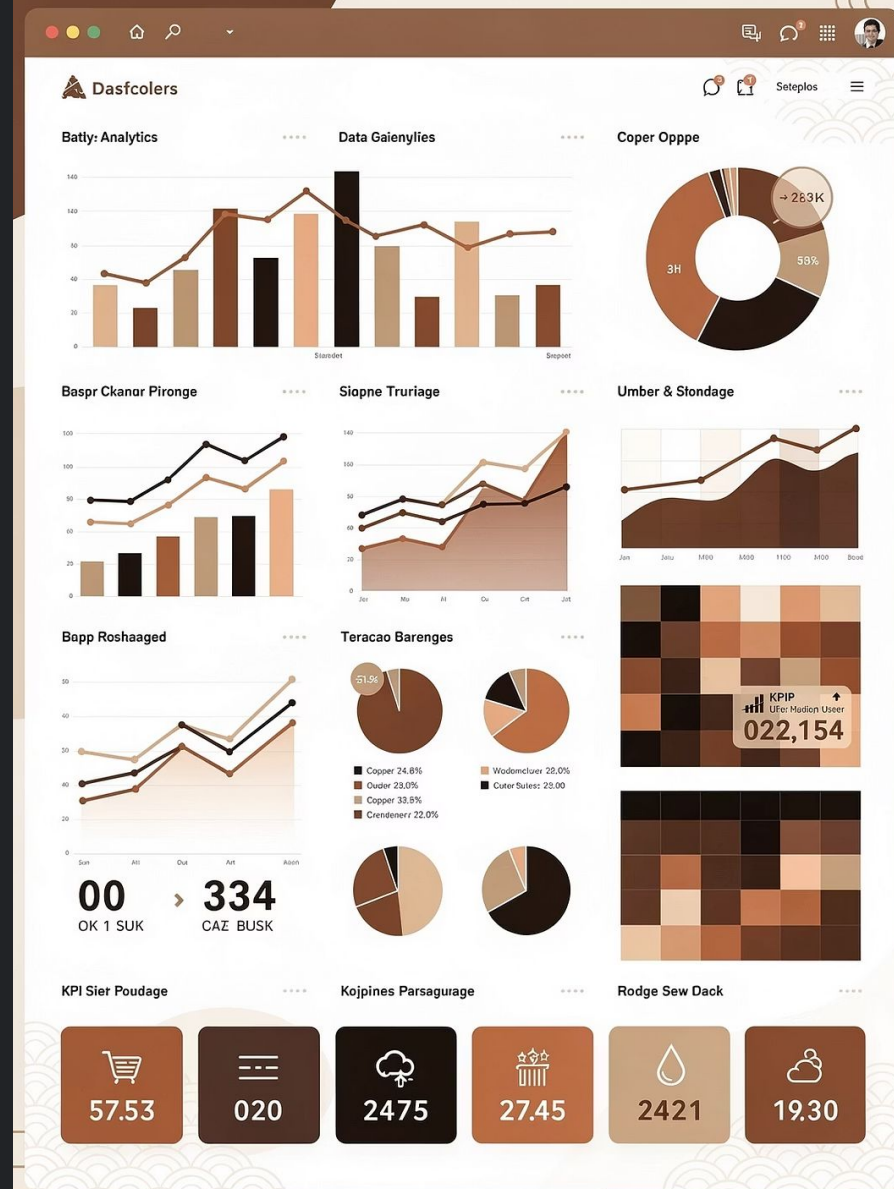


Yelp Analytics Dashboard

Transforming 8.6 Million Reviews into Actionable Business Intelligence

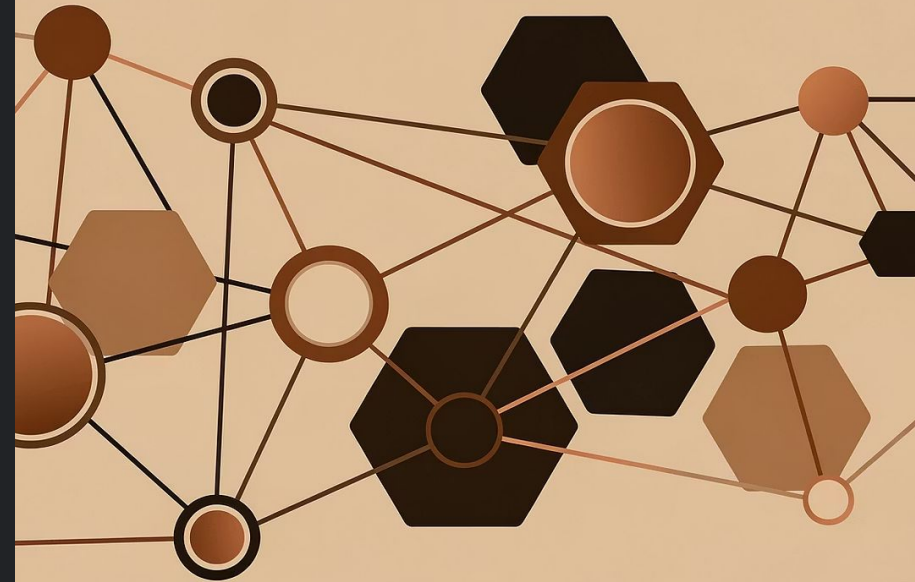
Lillian Wool, Eddie Steele, Alex Snyder



Project Purpose

Build a scalable analytics system that transforms 8.6 million+ Yelp reviews into actionable insights across four analytics tiers. The system supports real-world business decision-making by identifying operational issues, customer sentiment patterns, and data-driven strategic actions.

Implemented through a Polars-powered pipeline and Streamlit dashboard, this system delivers automated insights that help businesses understand what's happening, why it's happening, what's likely to happen next, and what actions should be taken.



Core Objectives

Efficient Processing

Process the full 8.6M-row Yelp dataset efficiently using Polars and lazy execution for optimal performance.

Four Analytics Tiers

Deliver automated insights across Descriptive, Diagnostic, Predictive, and Prescriptive analytics.

Interactive Dashboard

Build an interactive Streamlit dashboard for a unified, user-friendly experience.

Risk Detection

Detect business risks, sentiment trends, and performance shifts in real-time.

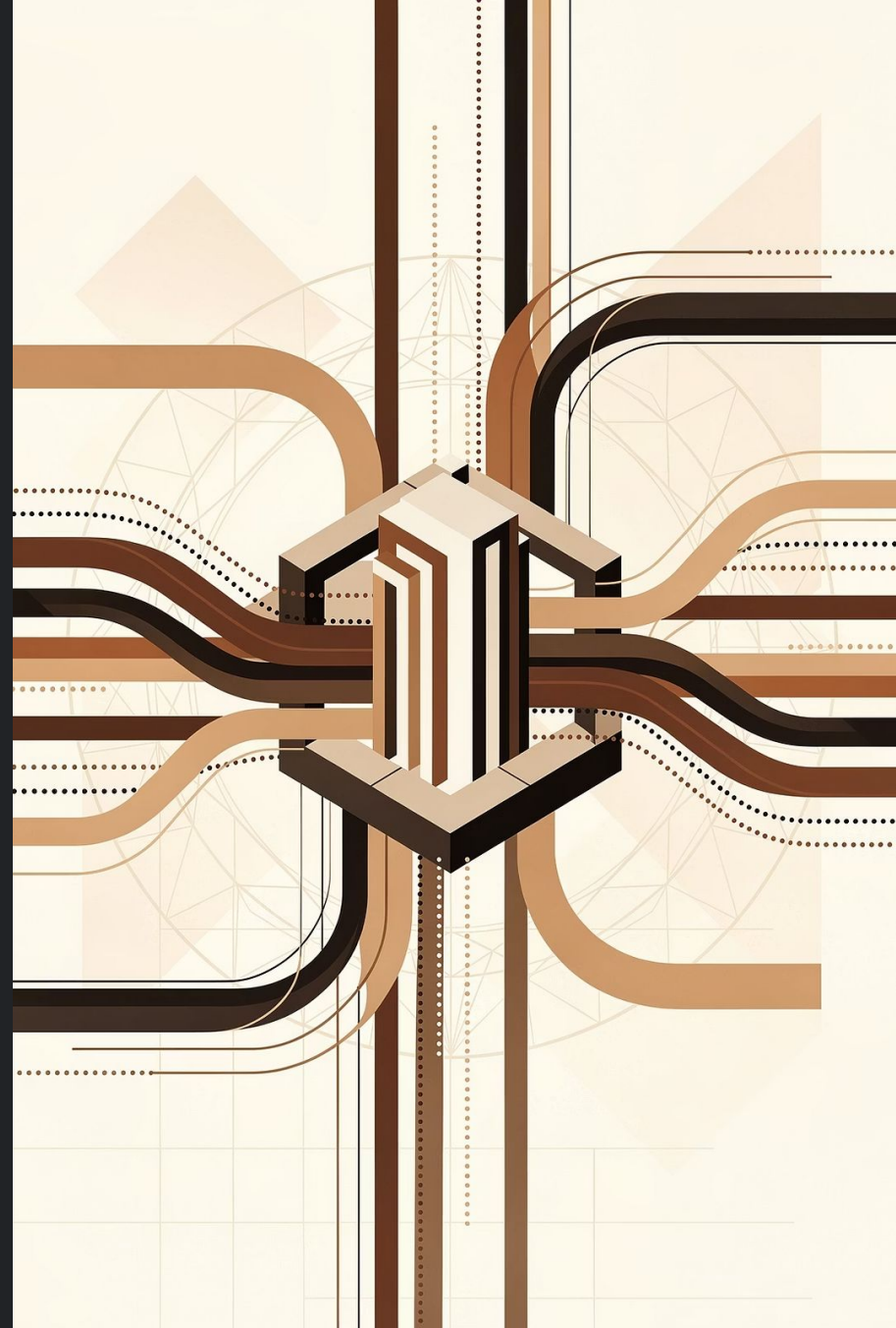
Data Inputs Powering the System

Core Data Sources

- Full Yelp review dataset (8.6M+ reviews)
- Yelp business metadata
- Cleaned review text (stripped + normalized)
- Time-stamped review data for trend detection

Advanced Features

- Emotion intensity features (anger, fear, joy, etc.)
- Linguistic structural features (noun %, adj %, verb %)
- Sentiment patterns and performance shifts



Dashboard Structure

Six comprehensive tabs provide a complete analytics experience

